

# Python Programming Basics: Study Guide

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## 1. Variables and Data Types

Variables aik container ki tarah hote hain jin mein aap data store karte hain. Python mein aapko variable ka type batane ki zaroorat nahi hoti, ye khud samajh jata hai. Ismein mukhtalif data types hoti hain jese integers (numbers), strings (text), aur floats (decimals). Jab aap koi value assign karte hain, to computer memory mein aik jagah reserved ho jati hai. Data types ka sahi istemal memory management ke liye bohat zaroori hai. Cybersecurity mein, variables ka use data packets ya logs ko store karne ke liye hota hai.

**Real-life Example:** Apne phone mein contact list save karna. Har contact ka naam (string) aur number (integer) aik variable hai jo memory mein save hota hai.

## 2. Conditional Statements (if, elif, else)

Conditional statements program ko faisla lene mein madad deti hain. Agar di gayi condition sahi (True) hoti hai, to code ka aik hissa chalta hai, warna doosra. Ye logic building ka basic pillar hai jahan program scenarios ke mutabiq react karta hai. `if` pehli condition check karta hai, `elif` agar pehli galat ho to doosri, aur `else` jab koi bhi condition meet na ho tab chalta hai. Cybersecurity mein firewall rules isi tarah kaam karte hain.

**Real-life Example:** Login system. Agar password sahi hai to "Access Granted", warna "Invalid Password" ka error show karna.

## 3. Loops (for and while)

Loops ka istemal tab hota hai jab aapne aik hi kaam ko baar baar repeat karna ho. `for` loop tab use hota hai jab aapko pata ho ke kitni dafa iteration karni hai, jese kisi list ke tamam items ko check karna. `while` loop tab chalta hai jab tak koi specific condition True rehti hai. Ye code ko chota aur efficient banata hai kyunke bar bar likhne ki zaroorat nahi padti. Automation mein loops ka role vital hai.

**Real-life Example:** Brute force attack mein password list ko aik-aik kar ke check karna loop ki aik misaal hai.

## 4. Functions

Functions code ka woh hissa hain jo aik specific task perform karne ke liye banaya jata hai. Inhein aap program mein kahin bhi bula (call) sakte hain, jis se bar bar code likhne ki mehnat bachti hai. Functions mein aap inputs (arguments) bhej sakte hain aur result wapas le sakte hain (return). Ye code ko organize aur readable banata hai. Bade software projects mein functions ke zariye code management asaan ho jati hai.

**Real-life Example:** Calculator ka button. Aap chahe 10 bar  $2+2$  karein, "equal to" ka function wahi logic baar baar apply kar raha hai.

## 5. Lists and Dictionaries

Lists aur Dictionaries data ko group karne ke tareeqe hain. `List` mein data aik sequence mein hota hai jise index ke zariye access kiya jata hai. `Dictionary` key-value pairs mein data rakhti hai, jese aik real dictionary mein word aur uska matlab hota hai. Python mein inka use bulk data handling ke liye kiya jata hai. Cybersecurity scripts mein logs ko organize karne ke liye ye best hain.

**Real-life Example:** List aapki grocery shopping ki list hai, aur Dictionary aapke friend ka phonebook hai jahan naam (key) ke badle number (value) save hai.